

Source-context $\mathcal{G}_s$

Option A: Source-available



Option B: Source-free



Pretrained
GNN
 F_Θ
(frozen)

Z_s^q

Z_t^q

Context gap extraction with queries

Shared trainable queries Q^q

Pool

Q^q

Pool

$c_{s,k}^q$

$c_{t,k}^q$

Context gap

$$d_k^q = c_{s,k}^q - c_{t,k}^q$$

Target local descriptor ζ_o^q

Local prompt generation

W^q
(trainable)

ϑ^q
(trainable)

Local relevance

$$a_{o,k}^q(\zeta_o^q, W^q)$$

Signed gates

$$\tanh(\vartheta_k^q)$$

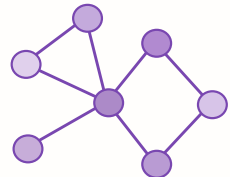
$$\text{Prompt: } p_o^q = \sum_k a_{o,k}^q \tanh(\vartheta_k^q) d_k^q$$

p_v^N

$p_{u \rightarrow v}^{M_\ell}$

Target graph prediction with prompted frozen GNN

Target graph G_t



+

$\text{MSG}_\Theta^{(1)}$

+

$\text{AGG/UPD}_\Theta^{(1)}$

...

$\text{AGG/UPD}_\Theta^{(L)}$

$\tilde{H}^{(L)}$

Readout

Prediction head

$\mathcal{G}_\psi$
(trainable)

$\mathcal{L}_{task}$